Manisha Tripathi

Summary:

- MS Quantitative Finance and Computer Science with strong experience in Model Validation, Quantitative analytics, derivatives pricing and market risk, market data, knowledge of FX, Equity and Fixed Income; and proficient in C++, Java, Excel VBA, Python, MATLAB, SQL, R, Bloomberg, Linux.

EXPERIENCE

CREDIT SUISSE, Singapore, Sr. Software Engineer, Fixed Income Analytics Jun 2015 – Feb 2016

- Part of the Fixed Income Front Office Support and Risk Technology team, supported valuation of Fixed Income products and their analytic calculations. Provided analytic support to the Credit Trading and Sales desk, and quants. Using pricing and risk analytics Excel VBA tool, Fixed Income SQL database, and Bloomberg to resolve Credit Trading and Sales analytics issues.
- Supported firm business needs in Fixed Income valuation and risk methodology.
- Transformed quantitative solutions into practical applications technically using Excel VBA
- Supported valuation and risk methods, and end-user tools compatible with firm's existing risk system using VBA
- Resolved live, often time-sensitive trader and sales desk issues, and coordinated with multiple teams like front office quants, reference product data, market data support, and asset control support team to get them addressed in an accurate and timely manner.

BARCLAYS CAPITAL, Singapore, Market Risk Analyst Feb 2013 – May 2013

- Part of Market Risk team, Responsible for management, monitoring and reporting of market risk VaR and position reporting for the Equities Futures and Options; Maintain risk data and VaR calculations, using historical simulation modelling using Excel VBA; Review issues in data to ensure that risk numbers are reported in an accurate and timely manner.
- Review of trade, risk and market data inputs on a daily basis via production and Validation of time series data using Bloomberg, Reuters, Asset Control; Setting of product risk margins and posting of intraday risk charges; Perform detailed investigations into back testing breaches and updating static data; Followed up with product data team if risk data are misaligned or missing; and gained an understanding of market risk for Equities, and identified potential risks.

ANZ BANKING GROUP, Singapore, Quantitative Developer Aug 2010 - Dec 2012

- Worked with the Front office FX Quantitative Applications team in Markets Asia division; responsible for development and review of derivatives pricing and risk analytics tool using C++, Excel VBA. Added the Vanna-Volga calibration to FX vanilla options, and developed strong working knowledge of FX options and structured products. Validated pricing models like Black Scholes, Heston, Local volatility models.
- Calculated Greeks in FX Basket Option Pricing. Worked on the payoff of FX TARN and priced them in Pegasus.
- Developed the FX volatility analysis tool, to serve as a post-check to validate volatility data that are fetched from Murex Cache and used in pricing. Displayed smile curve and implied spot density function by using GM library function. Checked whether there's an arbitrage opportunity, if using resulting volatility surface to price options.
- Validated and reviewed Black-Scholes pricing model for FX options, analysed and tested new features and modules in Basket and Compound option pricing and risk applications, and worked closely with sales and trading desk across Asia.
- Participated in workshops with stakeholders to gather requirements, Created FX Model Business Requirement Document, and Directed junior team members with usage and enhancement of FX pricing and risk framework.

BARKER INVESTMENT MANAGEMENT, Singapore, Portfolio Risk Intern July 2010

- Analysed portfolio exposures and help generate market revenues, using Excel VBA and SQL
- Used Bloomberg to analyse client data, manage portfolio risk, and actively participated in workshops with stakeholders to gather requirements, and created Business Requirement Document.

ACCENTURE TECHNOLOGY SOLUTIONS, Mumbai, Software Engineer Jun 2007–Mar 2009

- Developed Java application to standardise the process of selling financial products to customers. Modelled the workflow precisely to obtain information about clients to adhere to regulatory and corporate requirements for a product with its subsequent authentication and approvals; worked with the client team, and helped the projects meet the tight timelines with a quality delivery, received ASE achievement award.
- Participated in workshops with Client to gather requirements, created Business Requirement Document, Mapped solution, improvement, and enhancement; Performed UAT and ensured smooth rollout. Developed user and order management application, to be used by the client service centre where they can process the new users and their orders placed and remediate existing clients.

EDUCATION

WORLDQUANT UNIVERSITY, Louisiana, USA
Masters of Science in Quantitative Finance with Concentration in Alpha Design 2016 – Present

•**Courses**: Financial Markets, Statistics, Programming in Python, Algorithms, Econometrics, Alpha Design, Risk Management, Machine Learning, Capstone.

THE UNIVERSITY OF CHICAGO, Chicago, USA
Masters of Science in Financial Mathematics 2009 – 2010

•**Courses**: Mathematical Foundations of Option Pricing, Numerical Methods, Statistical Risk Management, Regression Analysis and Quantitative Trading Strategies, Portfolio Theory and Risk Management, Foreign Exchange, Fixed Income Derivatives, Introduction to Finance and Markets, Stochastic Calculus, Economics.

•**Project:** Developed the MATLAB regression model for Emerging Market Sovereign Credit Spreads, estimated the factors of the model like equity market returns, interest rates and FX rates, used the results in designing long short quantitative trading strategy based on regression analysis, and produced a model to capture the misprices between the credit markets and relevant market factors.

NATIONAL INSTITUTE OF TECHNOLOGY (NIT), Bhopal, India
Bachelor of Technology, Computer Science and Engineering 2003 – 2007

•**Courses**: Principles of Programming Languages, Software Programming; Artificial Intelligence, Theory of Computation, Data Structures, Analysis & Design of Algorithms, Engineering Physics, Engineering Chemistry, Mathematics, Environment Engineering, Communication Skills, Computer Graphics, Discrete Structures, Network Theory, Digital Electronics, Computer Architecture, Microprocessors, Database Management Systems, System Programming and Compiler Design, Operating Systems, Unix Shell Programming, Computer Networks, Data Warehousing & Mining, Wireless Communications & Mobile Computing, Embedded & Distributed System, Internet Technology, Neural Networks.

•**Project**: C++ Implementation of Network Address Translation – Protocol Translation from IPv4 to IPv6.